MECHTRON 2TA4 FINAL PROJECT: The Synabond

By: Prabhjot Bhandal, 400570830, bhandp9

The Synabond is a smart habit-tracking companion designed to help users build and maintain positive daily habits through interactive feedback. The system features a virtual pet whose mood and progression are directly influenced by the user's habit completion, creating a visual and engaging form of accountability.

The device is comprised of an ESP32-WROOM-32 MCU, a 128x64p OLED display, an RGB LED, an RTC DS3231 clock module and three pushbuttons to interact with the device. The device switches between two navigation states, an idle state and a log state. In the device's idle state, the user can see their virtual pet, its current level, its current mood, how many habits they have completed today and how much EXP their pet currently has. As the user completes their habits, their virtual pet gains experience to level up and evolve over time, which reinforces consistency in their habit completion through gamification. The pet's mood reflects the user's daily performance and provides immediate feedback through the OLED display and the colour of the RGB LED. In the device's log state, the user can view the current time given by the RTC DS3231 clock module as well as view and check off their habits. A user can change which habits they have throughout the day, and the tracker logs their completed habits at midnight and resets the pet's mood. The device also utilizes the ESP32's on-board EEPROM memory to store data and allow the virtual pet's information to persist, even when the device is off.

To enhance effectiveness, the Synabond incorporates a Hebbian learning model that analyzes patterns of habit completion and simulates the wiring of a person's neurons. The relationships between habits are stored in a weighted square matrix and the "keystone habit" (the habit with the strongest relationships to all other habits) is identified by having the largest sum of elements in its respective row in the matrix. After identifying the keystone habit, the system intelligently nudges the user to complete it as it will entice them to complete their other habits as well. These "nudges" are triggered at different times of day and will cause the RGB LED to flash and the nudged habit will be shown on the OLED screen. Upon every midnight logging, the relationships between habits are recomputed using different functions. If two habits are completed in a day, the growth rate is added to their relationship value, simulating neuroplasticity. This growth rate is based off a 2010 UCL study that found that it takes 66 days to form a habit [1].

$$\text{Current Relationship Value} += 0.05$$

All relationship values are multiplied by a decay rate, which simulates synaptic pruning. The decay rate was determined by basing it off the notion that it takes approximately three months to forget a habit [1].

$$\text{Decay Rate} = 0.99$$
$$\text{Current Relationship Value} *= \text{Decay Rate}$$

Finally, all relationship values are constrained between -1.0 and +1.0, with -1.0 signifying a completely decayed habit and +1.0 signifying a completely developed habit.

References

[1] P. Lally, “How habits are formed: Modelling habit formation in the real world,” Ph.D. dissertation, University College London, London, U.K., 2010.